NUVONIC

formerly Aquionics, Berson, Hanovia and Orca GmbH



PureLine ECO

OPTIMIZED UV TREATMENT FOR INDUSTRIAL & COMMERCIAL APPLICATIONS

Engineered with precision and driven by sustainability, ECO stands as a revolutionary UV system that promises to reshape the landscape of water treatment in the industrial & commercial markets

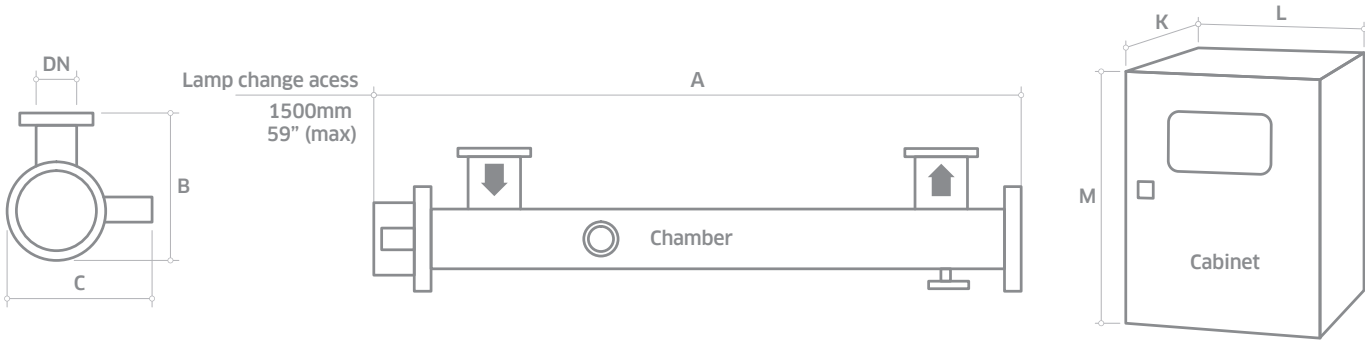
Efficiency: Reduce your carbon footprint significantly while maintaining the highest quality standards in your operations.

Compact Brilliance: ECO fits where others can't, delivering uncompromised results in limited space.

Optimized Performance: ECO is optimized for peak performance, providing consistently reliable water treatment solutions.

Application
Optimized UV for Food & Beverage

KEY FEATURES	WHAT IT GIVES YOU	BENEFITS FOR YOU
INTELLIGENCE		
UV intensity monitor measuring active wavelengths	Continuous verification of performance with in-built low intensity alarm	Easy to monitor and log system performance
OPTIMIZATION		
UV water treatment	Protect your process waters from microbiological contamination including chlorine resistant Cryptosporidium and Giardia	Does not affect taste and colour of final product
		No chemicals
Designed for the food and beverage industry	FDA-approved materials used for all wetted parts	Industry compliant materials
	Optional chamber with tri-clamp connections and <0.38 µm internal finish	Sanitary design
INTEGRATION		
Compact design	Can be fitted to skids	Easy integration
	Can be retrofitted to existing process	



MODEL NUMBER	FLOW** m³/h [gpm]	MAX POWER (W)	LAMP (QTY)	CHAMBER SIZE mm [in]					CHAMBER WEIGHT kg [lb]	
				A without wiper	A with wiper	B	C	DN	Chamber without wiper	Chamber with wiper
Pureline ECO 4	4.1 [18.1]	90	1	745 [29.3]	N/A	210 [8.3]	245 [9.5]	25 [1]	10.5 [23.1]	N/A
Pureline ECO 8	8.6 [37.9]	150	1	1245 [49]	N/A	230 [9.1]	255 [10.0]	50 [2]	18.5 [40.7]	N/A
Pureline ECO 15	17.8 [78.4]	250	1	1245 [49]	N/A	230 [9.1]	255 [10.0]	50 [2]	18.5 [40.7]	N/A
Pureline ECO 32	32 [140.9]	360	1	1785 [70.3]	N/A	230 [9.1]	275 [10.7]	80 [3]	24.3 [53.6]	N/A
Pureline ECO 40	41 [180.5]	380	1	1785 [70.3]	1895 [74.6]	335 [13.2]	305 [12.0]	80 [3]	42.7 [94.1]	49.0 [107.9]
Pureline ECO 85	86 [378.6]	750	2	1785 [70.3]	1895 [74.6]	335 [13.2]	320 [12.6]	100 [4]	42.7 [94.1]	49.0 [107.9]
Pureline ECO 125	125 [550.4]	1080	3	1785 [70.3]	1895 [74.6]	335 [13.2]	305 [12.0]	*125 [*5]	43.0 [94.8]	48.1 [106.0]
Pureline ECO 215	215 [946.6]	1420	4	1785 [70.3]	1895 [74.6]	455 [17.9]	355 [14.0]	150 [6]	66.4 [146.4]	73.7 [162.5]
Pureline ECO 225	225 [990.6]	1750	5	1785 [70.3]	1895 [74.6]	455 [17.9]	355 [14.0]	150 [6]	66.0 [145.5]	74.1 [163.4]
Pureline ECO 330	333 [1466.2]	1750	5	1790 [70.5]	1900 [74.8]	525 [20.7]	440 [17.3]	200 [8]	114.0 [251.3]	123.0 [271.2]
Pureline ECO 350	355 [1563.0]	2100	6	1790 [70.5]	1900 [74.8]	525 [20.7]	440 [17.3]	200 [8]	115.0 253.5]	124.0 [273.4]
Pureline ECO 380	385 [1695.1]	2440	7	1790 [70.5]	1900 [74.8]	525 [20.7]	440 [17.3]	200 [8]	116.0 [255.7]	125.0 [275.6]

*the DN is 150mm [6"] when connection type is Tri-clamp
** The disinfection capacity is based on 30mJ/cm2 at 99% (T10) UVT, end of lamp life
All dimensions are approximate for clearance purposes only. We have a policy of continuous product development, exact drawings are available on request.
All specifications are subject to change without notification. Your distributor or our account manager can advise on correct sizing and specification

UV CHAMBER	
Material:	StSt 316L / 1.4404
Process (Mating) connection:	PN10 to EN1092-1
End plate:	Removable plate (Tri-Clamp up to ECO 225, then flanged for larger models)
Drain connection:	Tri-Clamp 1" BS4825
Degree of protection:	IP54 equivalent to NEMA 12
Internal finish:	< 0.5 µm Ra (Welds ground out)
Lamp type:	Low Pressure Amalgam
Expected lamp life:	9000 hours
UV sensor:	Dry sensor
Temperature sensor	Yes
Seals:	EPDM, FDA 21 CFR 177.2600
Maximum CIP temperature:	95°C with cabinet electrically isolated
Working fluid temperature:	0-40°C
Operating pressure:	10 bar
Mounting feet	Included on ECO 215 and above

OPTIONS
Interconnecting cable lengths: 5 m
Chamber internal finish: < 0.38 µm Ra (Welds ground out)
Document support pack
Process (Mating) connection: Tri-Clamp BS4825 (for ECO 225 and smaller models)
ANSI B 16.5 Class 150 process (mating) connection
Carbon Steel Cabinet material
NEMA 4X (IP56 equivalent) cabinet
Wiper: Automatic (electrically driven) available on ECO 40 and above

MODEL NUMBER	LAMP (QTY)	STANDARD IP54, NEMA 12 CABINET SIZE mm [in]			OPTIONAL IP56, NEMA 4X CABINET SIZE * mm [in]			MAX. CABINET WEIGHT kg [lb]
		M	L	K	M	L	K	
Pureline ECO 4 - ECO 40	1 lamp systems	535 [21.1]	470 [18.5]	180 [7.1]	665 [26.2]	547 [21.5]	240 [9.4]	20.5 [45.2]
Pureline ECO 85 - ECO 125	2-3 lamp systems	600 [23.6]	490 [19.3}	210 [8.3]	750 [29.5]	557 [21.9]	270 [10.6]	29.1 [64.2]
Pureline ECO 215 - ECO 330	4-5 lamp systems	700 [27.6]	500 [19.7]	210 [8.3]	850 [33.4]	567 [22.3]	270 [10.6]	36.9 [81.4]
Pureline ECO 350 - ECO 380	6-7 lamp systems	800 [31.5]	510 [20.1]	210 [8.3]	950 [37.4]	567 [22.3]	270 [10.6]	45.9 [101.2]

*Stainless steel only
Allow dimension L in front of cabinet for door opening and panel access.
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CABINET (UV CONTROLLER)	
Material:	Stainless Steel 304
Controller:	UV Controller
HMI:	7" Touch screen
Interconnecting cable lengths:	5 m
IP rating:	IP54 / NEMA 12
Power supply:	220 V ± 10%, 50/60 Hz, single phase, L+N+G.
Operating temperature range:	5-40°C
Humidity:	< 90% noN-condensing
Control & Display:	Stepless power adjustment 50-100% (excluding ECO 4-8)
	All alarms and warnings
	Fixed dose running
	Water temperature
	Lamp running hours
	Flow rate (m^3/h or GPM)
	UV dose
	Lamp ready VFC
Alarm & Warning:	Low UV dose
	Lamp end pre-warning (time adjustable)
	Lamp fault alarm
	Chamber over temperature alarm
	UV sensor fault alarm
	Temperature sensor fault alarm
	Warning for lamp and quartz resetting
Input:	Flow 4-20 mA
	T_{10} 4-20 mA
	Remote On/Off
Output:	UV dose 4-20 mA
	Any system alarm VFC
	Any system warning VFC
	Low UV dose alarm VFC
	Lamp fault alarm VFC
	Lamp preheat VFC
	Lamp ready VFC

APPROVALS
CE, UKCA, UL, FCC

PureLine ECO

Also available in our ECO product range...



Breakdown and reduction of TOC



Reduce the bio-burden, protect against bio-fouling, lead to fewer CIP/SIP

Canada
+1 980 256 5700
americas@nuvoniuvc.com

China
+86 216 167 9599
apac@nuvonicuv.com

Germany
+44 175 351 5300
emea@nuvonicuv.com

Malaysia
+60 16 440 8834
sea@nuvonicuv.com



Mexico
+1 980 256 5700
americas@nuvonicuv.com

United Kingdom
+44 175 351 5300
emea@nuvonicuv.com

USA
+1 980 256 5700
americas@nuvonicuv.com

NUVONIC

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formerly Aquionics, Berson, Hanovia and Orca GmbH

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